

AWS Shared Responsibility Model

Security in the cloud and Security of the Cloud

Security OF the Cloud (AWS Responsibility): AWS is responsible for protecting the physical infrastructure that runs all cloud services. This includes the hardware, software, networking, and facilities (data centers) that host your applications. Think of AWS as the landlord who maintains the building.

Security IN the Cloud (Customer Responsibility): Customers are responsible for what they put into and configure within the cloud. This includes managing access controls, encrypting their own data, patching their operating systems, and configuring their applications securely. Think of the customer as the tenant who locks their own doors.

Visual Diagram: AWS Shared Responsibility Model

 AWS Responsibility (Security OF the Cloud)	 Shared Controls	 Customer Responsibility (Security IN the Cloud)
Physical server hardware	Patch Management	EC2 operating system patches
Data centre cooling & power	Configuration Management	S3 bucket access policy
Network firewall between AZs	Security Awareness Training	IAM user passwords & MFA
Hypervisor (virtualisation layer)		Encrypting data stored in S3
Database backups on RDS (managed)		Who can access your AWS account

		SSL certificate on your web app
		Database backups — MySQL on EC2

Categorisation Table

#	Item	Responsibility	Key Note
1	Physical server hardware	AWS	
2	EC2 operating system patches	Customer	
3	S3 bucket access policy	Customer	
4	Data centre cooling & power	AWS	
5	IAM user passwords & MFA	Customer	
6	Network firewall between AZs	AWS	
7	Encrypting data stored in S3	Customer	

8	Who can access your AWS account	Customer	
9	SSL certificate on your web app	Customer	
10	Hypervisor (virtualisation layer)	AWS	
11	Database backups on RDS (managed)	AWS	<i>AWS manages the engine</i>
12	Database backups — MySQL on EC2	Customer	<i>You manage everything on EC2</i>

Explanation (for a Lagos Restaurant Owner)

What AWS handles: When you move your restaurant's POS system and customer data to AWS, Amazon takes full responsibility for the physical computers, the building where they are stored, the electricity, the cooling, and the basic plumbing that makes the internet work. You never need to worry about a server catching fire or a hard drive failing.

What you still handle: However, just like renting a shop space doesn't mean the landlord locks your safe for you, AWS does not manage who can log in to your account, how your customer records are protected, or whether your staff are using strong passwords. Those decisions remain entirely yours.

Why it matters for your business: This matters because if a hacker steals your customers' payment details because you left your cloud storage wide open, AWS is not legally or financially responsible, you are. Which is why understanding this split before you migrate is one of the most important steps you can take to protect your business and your customers' trust.

BONUS CHALLENGE — 2019 Capital One Data Breach Analysis

2019 Capital One Data Breach — Shared Responsibility Analysis

Verdict: Customer misconfiguration — NOT an AWS failure.

What happened: A former AWS employee exploited a misconfigured Web Application Firewall (WAF) on Capital One's AWS environment to access ~100 million customer records.

Customer failure #1: Overly permissive IAM role: the WAF had an IAM role with excessive permissions (metadata service access), violating the principle of least privilege.

Customer failure #2: Misconfigured security group and WAF rules allowed SSRF (Server-Side Request Forgery) attacks to succeed — a configuration the customer controlled.

Customer failure #3: Lack of monitoring and alerting: Capital One failed to detect abnormal data access patterns in S3, a customer responsibility under the shared model.

AWS's role: AWS provided the tools (IAM, WAF, CloudTrail, GuardDuty) to prevent and detect the breach. Capital One did not use them correctly — the infrastructure itself was not compromised.

Day 3 of AWS Cloud training: understanding the Shared Responsibility Model. Key insight — moving to the cloud does NOT mean zero security responsibility. AWS secures the physical infrastructure. YOU are responsible for your data, access controls, and application security. Here's my visual summary 📌 [attach diagram] #AWS #CloudSecurity #LearningInPublic #BaseStackAcademy